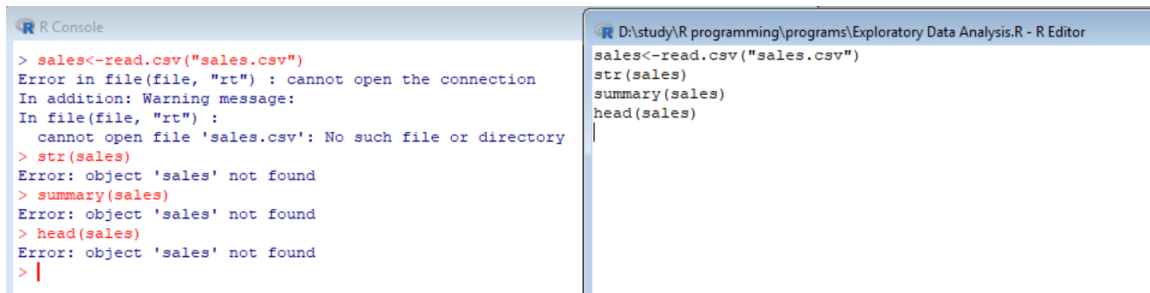


Case Study R Programs for Exploratory Data Analysis (EDA)

1. Exploratory Data Analysis

Case Study: Case: Analyze a supermarket sales dataset to identify trends before prediction.

```
sales<-read.csv("sales.csv")
str(sales)
summary(sales)
head(sales)
```



The screenshot shows two windows from an R environment. The left window, titled 'R Console', displays the following error messages:

```
> sales<-read.csv("sales.csv")
Error in file(file, "rt") : cannot open the connection
In addition: Warning message:
In file(file, "rt") :
  cannot open file 'sales.csv': No such file or directory
> str(sales)
Error: object 'sales' not found
> summary(sales)
Error: object 'sales' not found
> head(sales)
Error: object 'sales' not found
> |
```

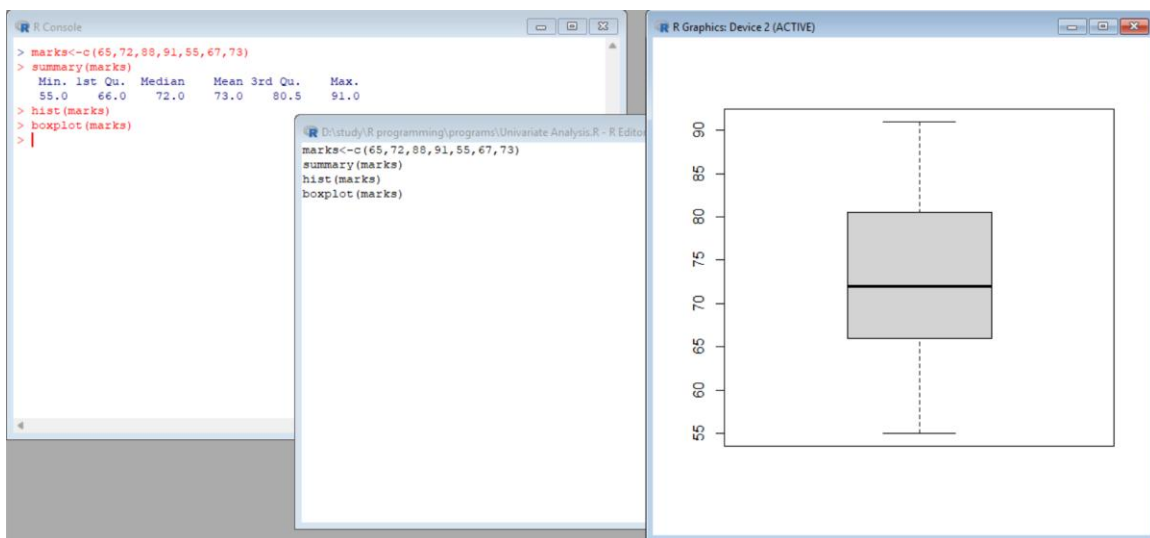
The right window, titled 'D:\study\R programming\programs\Exploratory Data Analysis.R - R Editor', shows the same R code being executed:

```
sales<-read.csv("sales.csv")
str(sales)
summary(sales)
head(sales)
```

2. Univariate Analysis

Case Study: Case: Analyze distribution of student marks.

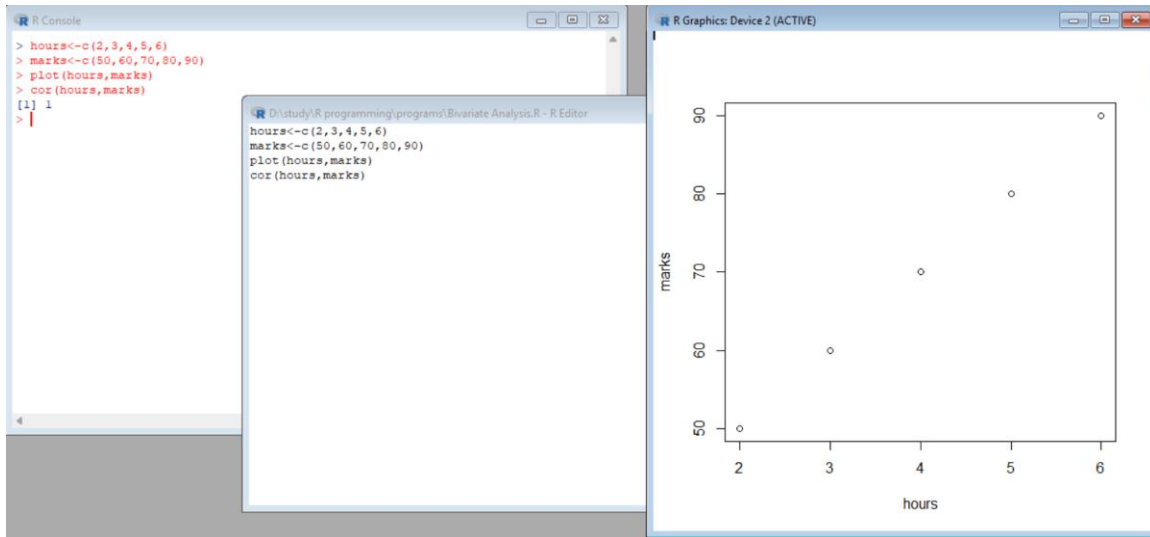
```
marks<-c(65,72,88,91,55,67,73)
summary(marks)
hist(marks)
boxplot(marks)
```



3. Bivariate Analysis

Case Study: Case: Study relationship between study hours and marks.

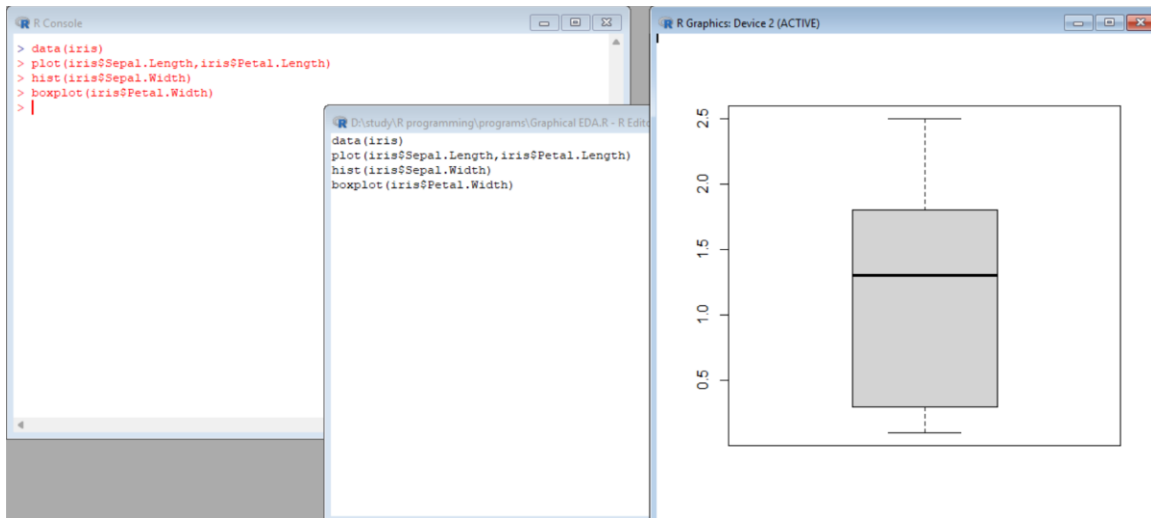
```
hours<-c(2,3,4,5,6)
marks<-c(50,60,70,80,90)
plot(hours,marks)
cor(hours,marks)
```



4. Graphical EDA

Case Study: Case: Visualize iris dataset.

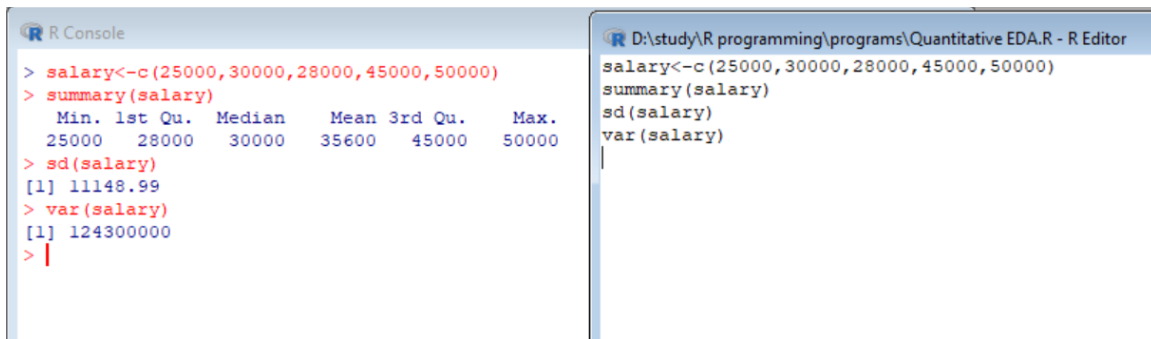
```
data(iris)
plot(iris$Sepal.Length,iris$Petal.Length)
hist(iris$Sepal.Width)
boxplot(iris$Petal.Width)
```



5. Quantitative EDA

Case Study: Case: Summarize employee salaries.

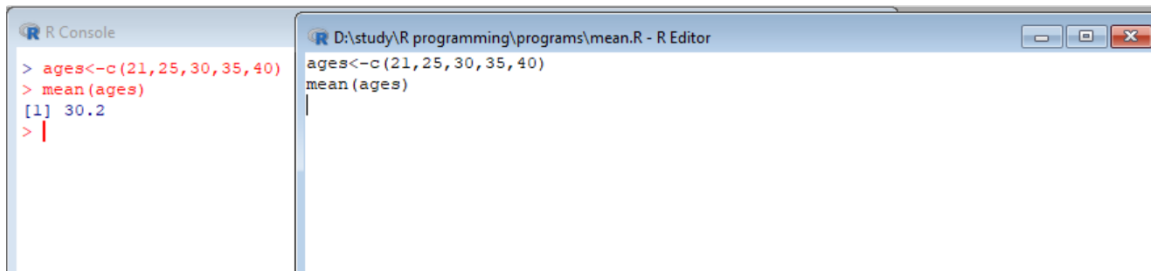
```
salary<-c(25000,30000,28000,45000,50000)
summary(salary)
sd(salary)
var(salary)
```



6. Mean

Case Study: Hospital patient ages.

```
ages<-c(21,25,30,35,40)
mean(ages)
```



The screenshot shows two windows from an R environment. The 'R Console' window on the left contains the following code and output:

```
> ages<-c(21,25,30,35,40)
> mean(ages)
[1] 30.2
> |
```

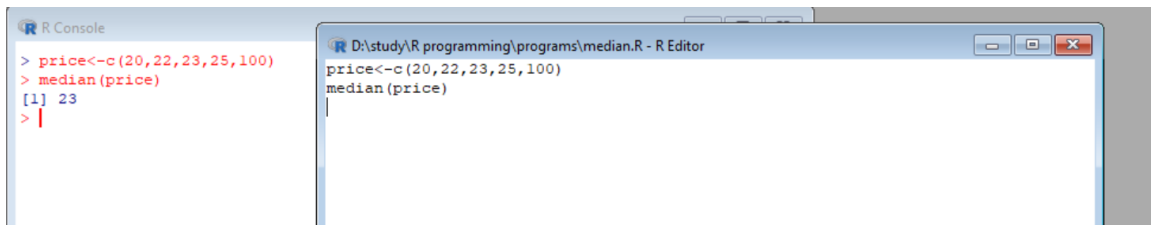
The 'R Editor' window on the right, titled 'D:\study\R programming\programs\mean.R - R Editor', contains the same code:

```
ages<-c(21,25,30,35,40)
mean(ages)
```

7. Median

Case Study: House prices.

```
price<-c(20,22,23,25,100)
median(price)
```



The screenshot shows two windows from an R environment. The 'R Console' window on the left contains the following code and output:

```
> price<-c(20,22,23,25,100)
> median(price)
[1] 23
> |
```

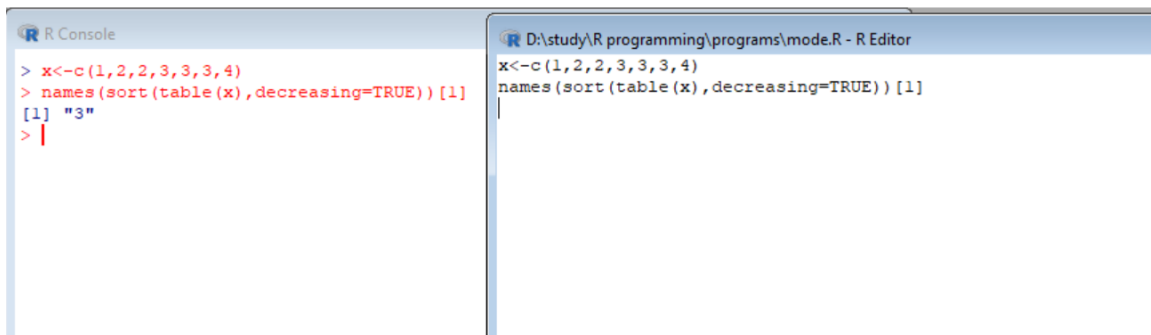
The 'R Editor' window on the right, titled 'D:\study\R programming\programs\median.R - R Editor', contains the same code:

```
price<-c(20,22,23,25,100)
median(price)
```

8. Mode

Case Study: Most purchased product.

```
x<-c(1,2,2,3,3,3,4)
names(sort(table(x),decreasing=TRUE))[1]
```



The screenshot shows two windows from an R environment. The 'R Console' window on the left contains the following code and output:

```
> x<-c(1,2,2,3,3,3,4)
> names(sort(table(x),decreasing=TRUE))[1]
[1] "3"
> |
```

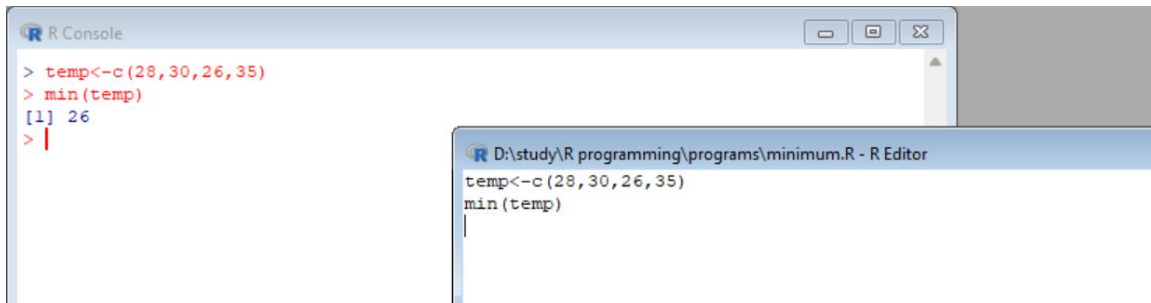
The 'R Editor' window on the right, titled 'D:\study\R programming\programs\mode.R - R Editor', contains the same code:

```
x<-c(1,2,2,3,3,3,4)
names(sort(table(x),decreasing=TRUE))[1]
```

9. Minimum

Case Study: Lowest temperature.

```
temp<-c(28,30,26,35)
min(temp)
```



The screenshot shows two windows from an R environment. The 'R Console' window on the left contains the following code and output:

```
> temp<-c(28,30,26,35)
> min(temp)
[1] 26
> |
```

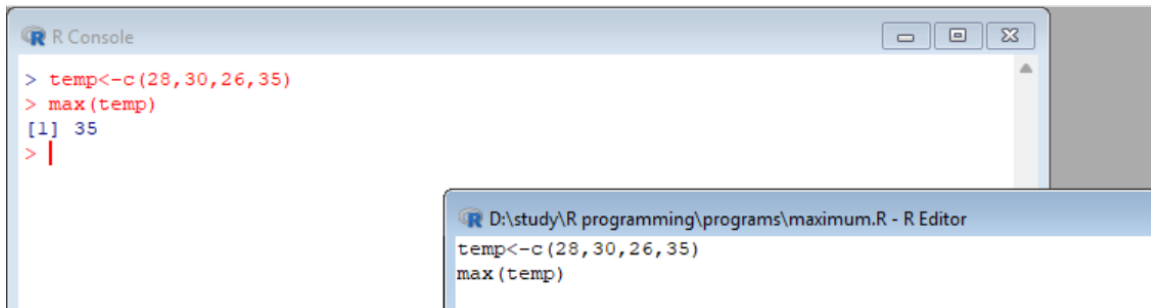
The 'R Editor' window on the right, titled 'D:\study\R programming\programs\minimum.R - R Editor', contains the following code:

```
temp<-c(28,30,26,35)
min(temp)
|
```

10. Maximum

Case Study: Highest temperature.

```
temp<-c(28,30,26,35)
max(temp)
```



The screenshot shows two windows from an R environment. The 'R Console' window on the left contains the following code and output:

```
> temp<-c(28,30,26,35)
> max(temp)
[1] 35
> |
```

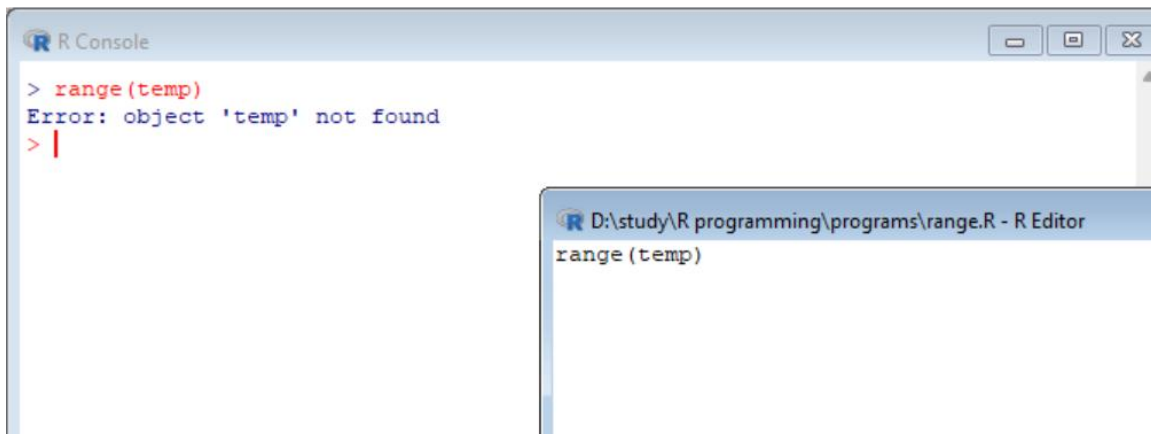
The 'R Editor' window on the right, titled 'D:\study\R programming\programs\maximum.R - R Editor', contains the following code:

```
temp<-c(28,30,26,35)
max(temp)
```

11. Range

Case Study: Temperature variation.

```
range(temp)
```



The screenshot shows two windows from an R environment. The 'R Console' window on the left contains the following code and output:

```
> range(temp)
Error: object 'temp' not found
> |
```

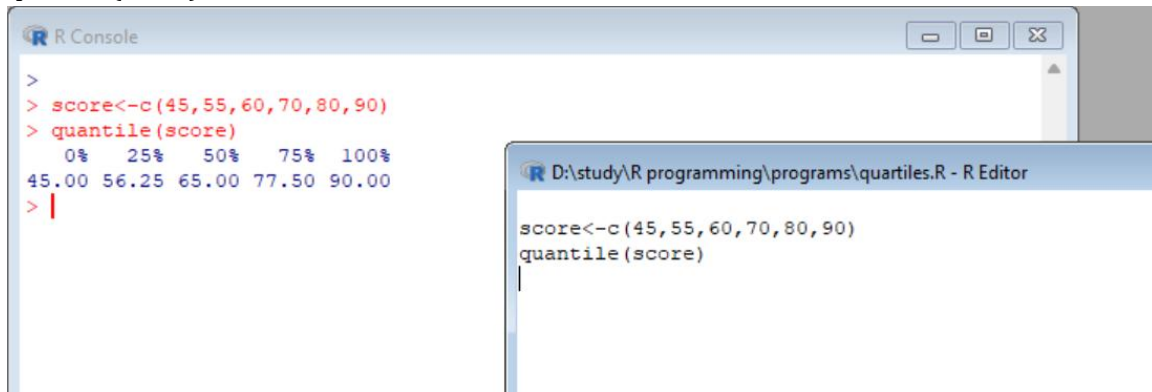
The 'R Editor' window on the right, titled 'D:\study\R programming\programs\range.R - R Editor', contains the following code:

```
range(temp)
```

12. Quartiles

Case Study: Exam score spread.

```
score<-c(45,55,60,70,80,90)
quantile(score)
```



The screenshot shows two windows from an R environment. The 'R Console' window on the left displays the execution of the code: `score<-c(45,55,60,70,80,90)` and `quantile(score)`. The output shows the quantiles at 0%, 25%, 50%, 75%, and 100%: 45.00, 56.25, 65.00, 77.50, and 90.00. The 'R Editor' window on the right, titled 'D:\study\R programming\programs\quartiles.R - R Editor', contains the same code that was executed in the console.

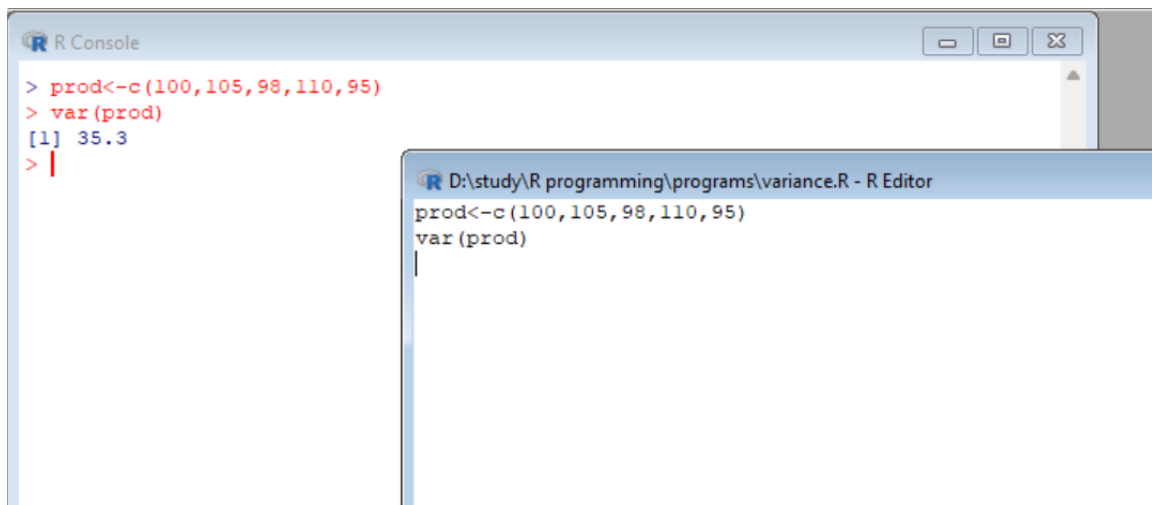
```
R Console
>
> score<-c(45,55,60,70,80,90)
> quantile(score)
 0%   25%   50%   75%  100%
45.00 56.25 65.00 77.50 90.00
> |

D:\study\R programming\programs\quartiles.R - R Editor
score<-c(45,55,60,70,80,90)
quantile(score)
|
```

13. Variance

Case Study: Production consistency.

```
prod<-c(100,105,98,110,95)
var(prod)
```



The screenshot shows two windows from an R environment. The 'R Console' window on the left displays the execution of the code: `prod<-c(100,105,98,110,95)` and `var(prod)`. The output is `[1] 35.3`. The 'R Editor' window on the right, titled 'D:\study\R programming\programs\variance.R - R Editor', contains the same code that was executed in the console.

```
R Console
> prod<-c(100,105,98,110,95)
> var(prod)
[1] 35.3
> |

D:\study\R programming\programs\variance.R - R Editor
prod<-c(100,105,98,110,95)
var(prod)
|
```